

Modul Praktikum

Pengembangan Aplikasi Basisdata

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MODUL PRAKTIKUM 14

Grafik, BLOB Gambar, Import Data Excel, dan Deploy Aplikasi

A. Tujuan Praktikum

1. Menampilkan Grafik.
2. Menampilkan Gambar dan Menginput Gambar ke dalam Database.
3. Upload Data dari File Excel ke dalam Database.
4. Mendeploy Aplikasi.

B. Alat dan Bahan

- SQL Server
- SQL Server Management Studio (SSMS)
- Visual Studio (Minimal 2019 dengan .NET Framework)
- C# Windows Forms (WinForms)

C. Langkah-Langkah Praktikum

Setup Awal

1. Instal terlebih dahulu **Inno Setup Compiler**.
2. Tambahkan Kolom Foto pada Tabel Mahasiswa.

```
alter table mahasiswa  
add foto varbinary(max);
```

3. Ubah Stored Procedure Read, Insert, dan Update Mahasiswa
 - a. Stored Procedure Read

```
ALTER PROCEDURE sp_GetMahasiswa  
AS  
BEGIN  
    SELECT  
        mahasiswa.NIM as NIM,  
        mahasiswa>Nama as Nama,  
        mahasiswa.JenisKelamin as JenisKelamin,  
        mahasiswa.TanggalLahir as TanggalLahir,  
        mahasiswa.Alamat as Alamat,  
        mahasiswa.Foto as Foto,  
        ProgramStudi>NamaProdi as NamaProdi  
    FROM  
        Mahasiswa  
    JOIN  
        ProgramStudi  
    ON mahasiswa.KodeProdi = ProgramStudi.KodeProdi;  
END
```

```

ALTER PROCEDURE sp_GetMahasiswaByNIM
@pNIM char(11)
AS
BEGIN
    SELECT
        mahasiswa.NIM,
        mahasiswa>Nama,
        mahasiswa.JenisKelamin,
        mahasiswa.TanggalLahir,
        mahasiswa.Alatnat,
        mahasiswa.Foto as Foto,
        ProgramStudi>NamaProdi
    FROM
        Mahasiswa
    JOIN
        ProgramStudi
    ON mahasiswa.KodeProdi = ProgramStudi.KodeProdi
    WHERE mahasiswa.NIM = @pNIM;
END

```

b. Stored Procedure Insert

```

ALTER PROCEDURE sp_InsertMahasiswa
@pNIM char(11),
@pNama varchar(100),
@pAlamat varchar(200),
@pJenisKelamin char(1),
@pTanggalLahir datetime,
@pKodeProdi char(4),
@pFoto varbinary(max)
AS
BEGIN
    INSERT INTO Mahasiswa
        (NIM, Nama, Alamat, JenisKelamin, TanggalLahir, KodeProdi, TanggalDaftar, foto)
    VALUES
        (@pNIM, @pNama, @pAlamat, @pJenisKelamin, @pTanggalLahir, @pKodeProdi, GETDATE(), @pFoto);
END

```

c. Stored Procedure Update

```

ALTER PROCEDURE sp_UpdateMahasiswa
@pNIM char(11),
@pNama varchar(100),
@pAlamat varchar(200),
@pJenisKelamin char(1),
@pTanggalLahir datetime,
@pKodeProdi char(4),
@pFoto varbinary(max)
AS
BEGIN
    UPDATE Mahasiswa
        SET Nama = @pNama, Alamat = @pAlamat, JenisKelamin = @pJenisKelamin,
            TanggalLahir = @pTanggalLahir, KodeProdi = @pKodeProdi, foto = @pFoto
        WHERE NIM = @pNIM;
END

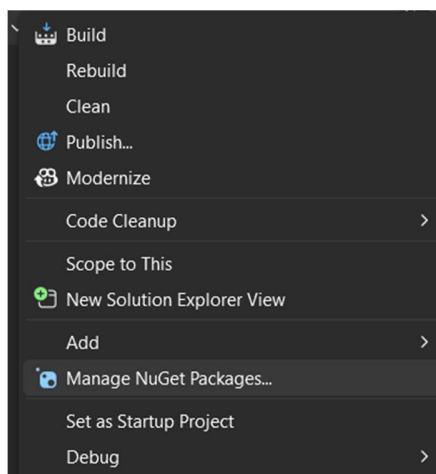
```

4. Tambahkan Stored Procedure untuk Chart

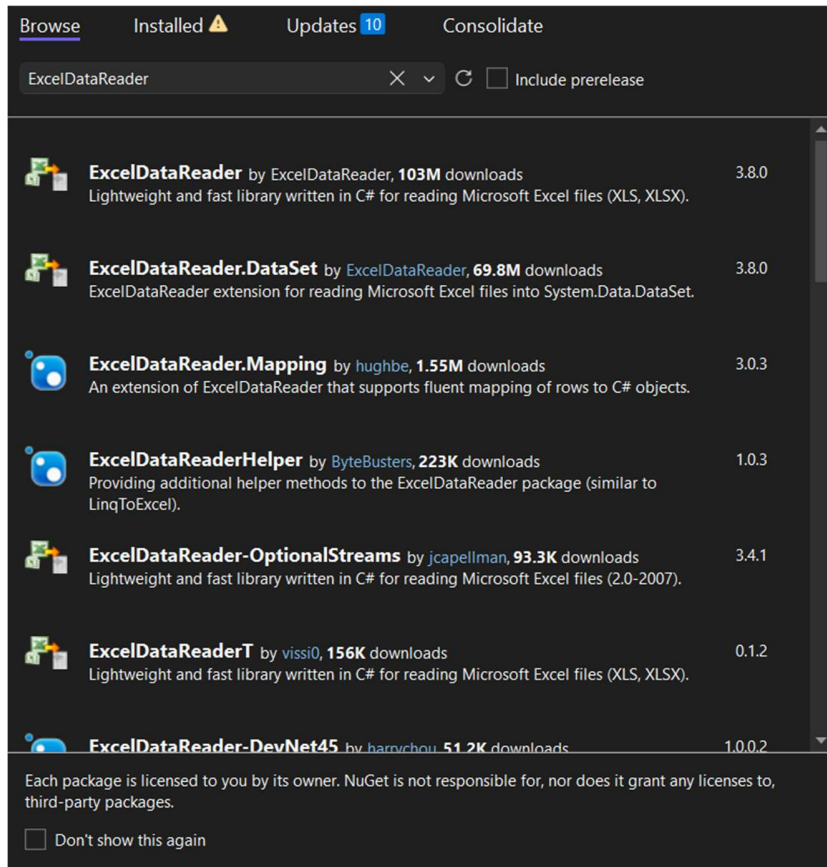
```
CREATE PROCEDURE sp_DashBoard
AS
BEGIN
    SELECT
        ProgramStudi>NamaProdi as>NamaProdi,
        count(mahasiswa.KodeProdi) as>JmlhMhs
    FROM
        Mahasiswa
    JOIN
        ProgramStudi
    ON mahasiswa.KodeProdi = ProgramStudi.KodeProdi
    GROUP BY ProgramStudi>NamaProdi;
END

CREATE PROCEDURE sp_DashBoardByTahun
    @inTglMsuk char(4)
AS
BEGIN
    SELECT
        ProgramStudi>NamaProdi as>NamaProdi,
        count(mahasiswa.KodeProdi) as>JmlhMhs
    FROM
        Mahasiswa
    JOIN
        ProgramStudi
    ON mahasiswa.KodeProdi = ProgramStudi.KodeProdi
    WHERE YEAR(mahasiswa.TanggalDaftar) = @inTglMsuk
    GROUP BY ProgramStudi>NamaProdi;
END
```

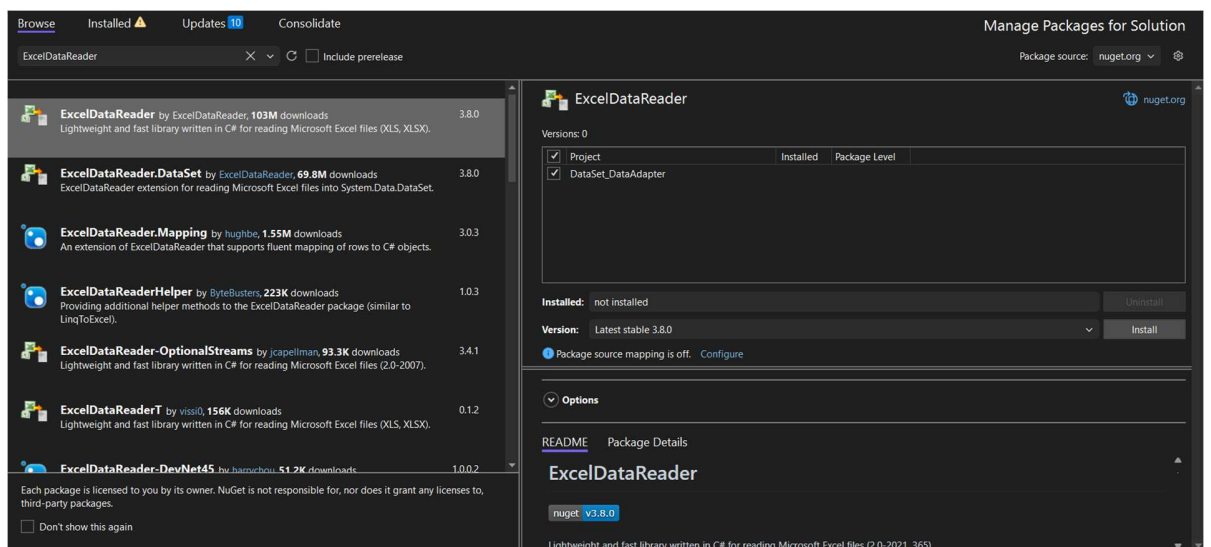
5. Tambahkan Package ExcelDataReader untuk membaca data dari Excel kedalam Project
- Klik kanan pada proyek di Solution Explorer.
 - Pilih “Manage NuGet Packages...” dari menu konteks.



- c. Di jendela NuGet Package Manager, pilih tab "Browse".
- d. Ketik "ExcelDataReader" di kotak pencarian.

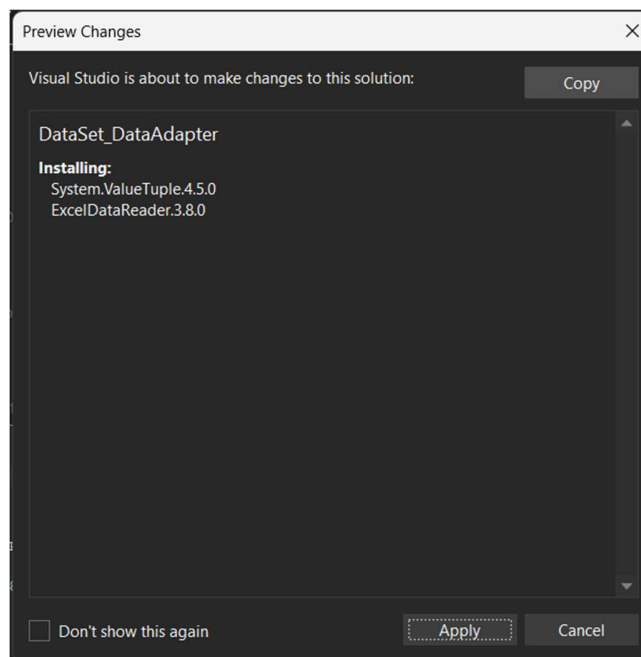


- e. Dari hasil pencarian, pilih "ExcelDataReader".
- f. Klik tombol "Install" dan ikuti prompt untuk menyelesaikan instalasi.



- g. Visual Studio mungkin meminta untuk meninjau perubahan dan menerima lisensi paket.

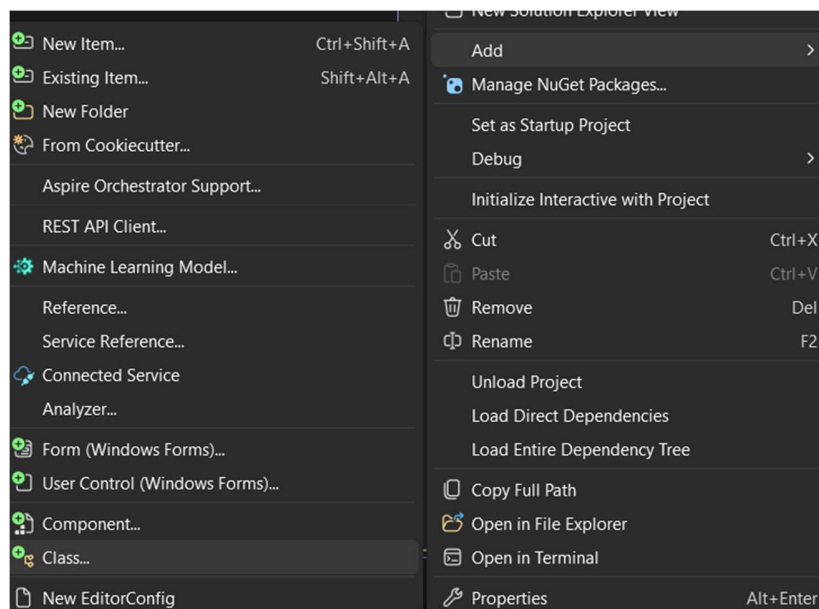
- h. Klik "OK" atau "I Accept" atau "Apply" untuk menyelesaikan proses instalasi.



- i. Jika instalasi ExcelDataReader selesai, lakukan langkah f - h untuk ExcelDataReader.DataSet.

Modifikasi Aplikasi

6. Buat Class Baru, dengan untuk Database Arsitektur Logic
- Klik kanan pada nama project
 - Pilih Add, kemudian klik Class..



- c. Beri nama DAL.

- d. Pindahkan code terkait logic database dari Form yang telah ada kedalam DAL

```
static string connectionString = "Data Source=NamaServer;Initial Catalog=DBAKademikADO;User ID=sa;Password=PasswordSA;";  
1 reference  
public string GetConnectionString()  
{  
    return connectionString;  
}  
  
SqlConnection conn = new SqlConnection(connectionString);  
  
SqlDataAdapter da;  
DataTable dtMahasiswa;  
DataTable dtProdi;  
2 references  
public int CountMhs()  
{  
    if (conn.State == ConnectionState.Closed)  
    {  
        conn.Open();  
    }  
  
    SqlCommand cmd = new SqlCommand("sp_CountMahasiswa", conn);  
    cmd.CommandType = CommandType.StoredProcedure;  
  
    SqlParameter outputParam = new SqlParameter("@pCount", SqlDbType.Int);  
    outputParam.Direction = ParameterDirection.Output;  
  
    cmd.Parameters.Add(outputParam);  
  
    cmd.ExecuteNonQuery();  
  
    return Convert.ToInt32(outputParam.Value);  
}  
1 reference  
public DataTable GetMhs()  
{  
    if (conn.State == ConnectionState.Closed)  
    {  
        conn.Open();  
    }  
  
    SqlCommand cmd = new SqlCommand("sp_GetMahasiswa", conn);  
    cmd.CommandType = CommandType.StoredProcedure;  
  
    da = new SqlDataAdapter(cmd);  
  
    dtMahasiswa = new DataTable();  
    da.Fill(dtMahasiswa);  
  
    return dtMahasiswa;  
}
```

```

public void InsertMhs(string nim, string nama, string alamat, string jenisKelamin, DateTime tanggalLahir, string kodeProdi, byte[] foto)
{
    if (conn.State == ConnectionState.Closed)
    {
        conn.Open();
    }
    SqlTransaction trans = conn.BeginTransaction();
    try
    {
        SqlCommand command = new SqlCommand("sp_InsertMahasiswa", conn);
        command.CommandType = CommandType.StoredProcedure;

        command.Parameters.AddWithValue("pNIM", nim);
        command.Parameters.AddWithValue("pNama", nama);
        command.Parameters.AddWithValue("pAlamat", alamat);
        command.Parameters.AddWithValue("pTanggalLahir", tanggalLahir);
        command.Parameters.AddWithValue("pJenisKelamin", jenisKelamin);
        command.Parameters.AddWithValue("pNmProdi", kodeProdi);
        command.Parameters.AddWithValue("pFoto", foto);

        command.ExecuteNonQuery();
        trans.Commit();
    }
    catch (Exception ex)
    {
        trans.Rollback();
    }
    finally
    {
        conn.Close();
    }
}

public void UpdateMhs(string nim, string nama, string alamat, string jenisKelamin, DateTime tanggalLahir, string kodeProdi, byte[] foto)
{
    if (conn.State == ConnectionState.Closed)
    {
        conn.Open();
    }

    SqlCommand command = new SqlCommand("sp_UpdateMahasiswa", conn);

    command.Parameters.AddWithValue("pNIM", nim);
    command.Parameters.AddWithValue("pNama", nama);
    command.Parameters.AddWithValue("pAlamat", alamat);
    command.Parameters.AddWithValue("pJenisKelamin", jenisKelamin);
    command.Parameters.AddWithValue("pTanggalLahir", tanggalLahir);
    command.Parameters.AddWithValue("pNmProdi", kodeProdi);
    command.Parameters.AddWithValue("pFoto", foto);

    command.CommandType = CommandType.StoredProcedure;

    command.ExecuteNonQuery();
}

public void DeleteMhs(string nim)
{
    if (conn.State == ConnectionState.Closed)
    {
        conn.Open();
    }

    SqlCommand cmd = new SqlCommand("sp_DeleteMahasiswa", conn);
    cmd.Parameters.AddWithValue("pNIM", nim);
    cmd.CommandType = CommandType.StoredProcedure;

    cmd.ExecuteNonQuery();
}

```



```

public void resetData()
{
    if (conn.State == ConnectionState.Closed)
    {
        conn.Open();
    }

    string deleteQuery = "DELETE FROM mahasiswa;";
    SqlCommand cmdDelete = new SqlCommand(deleteQuery, conn);
    cmdDelete.ExecuteNonQuery();

    string insertQuery = @"
        INSERT INTO mahasiswa
        SELECT * FROM mahasiswa_backup;";
    SqlCommand cmdInsert = new SqlCommand(insertQuery, conn);
    cmdInsert.ExecuteNonQuery();
}

```

1 reference

```

public void testInject(string nim)
{
    if (conn.State == ConnectionState.Closed)
    {
        conn.Open();
    }

    string query = "Update mahasiswa set nama = 'HACKED' where NIM = " + nim;
    SqlCommand cmd = new SqlCommand(query, conn);
    cmd.ExecuteNonQuery();
}

```

```

public DataTable GetMhsByNIM(string nim)
{
    if (conn.State == ConnectionState.Closed)
    {
        conn.Open();
    }

    SqlCommand cmd = new SqlCommand("sp_GetMahasiswaByNIM", conn);
    cmd.CommandType = CommandType.StoredProcedure;

    cmd.Parameters.AddWithValue("pNIM", nim);
    da = new SqlDataAdapter(cmd);

    dtMahasiswa = new DataTable();
    da.Fill(dtMahasiswa);

    return dtMahasiswa;
}

```

1 reference

```

public void InsertLog(string message)
{
    if (conn.State == ConnectionState.Closed)
    {
        conn.Open();
    }

    SqlCommand cmd = new SqlCommand("sp_LogMessage", conn);

    cmd.Parameters.AddWithValue("psn", message);
    cmd.CommandType = CommandType.StoredProcedure;
    cmd.ExecuteNonQuery();
}

```

```

public DataTable getProdi()
{
    if (conn.State == ConnectionState.Closed)
    {
        conn.Open();
    }

    SqlCommand cmd = new SqlCommand("select namaprodi from prodi", conn);
    cmd.CommandType = CommandType.Text;
    dtProdi = new DataTable();
    da = new SqlDataAdapter(cmd);
    da.Fill(dtProdi);

    return dtProdi;
}

2 references
public DataTable getDataRekap(string prodi, DateTime tanggalMasuk)
{
    if (conn.State == ConnectionState.Closed)
    {
        conn.Open();
    }

    SqlCommand cmd = new SqlCommand("sp_Report", conn);
    cmd.CommandType = CommandType.StoredProcedure;
    cmd.Parameters.AddWithValue("@inProdi", prodi);
    cmd.Parameters.AddWithValue("@inTglMsuk", tanggalMasuk.Year.ToString());

    da = new SqlDataAdapter(cmd);

    dtMahasiswa = new DataTable();
    da.Fill(dtMahasiswa);
    return dtMahasiswa;
}

public DataTable getAllDataChart()
{
    if (conn.State == ConnectionState.Closed)
    {
        conn.Open();
    }

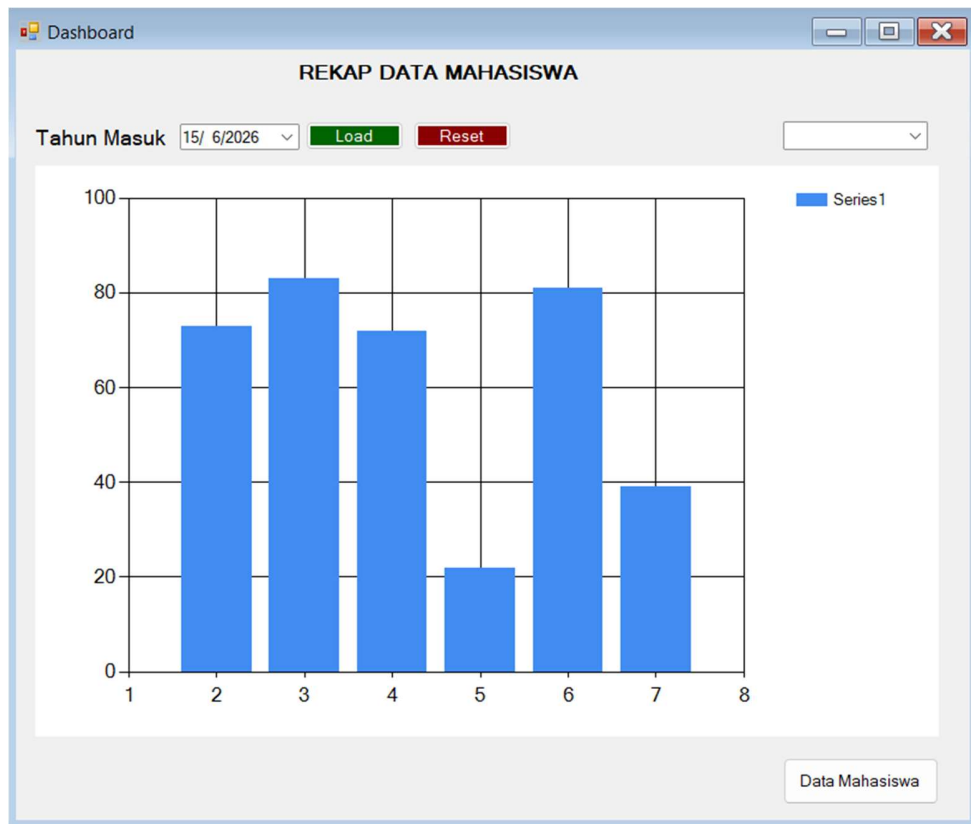
    SqlCommand cmd = new SqlCommand("sp_DashBoard", conn);
    cmd.CommandType = CommandType.StoredProcedure;
    da = new SqlDataAdapter(cmd);
    dtMahasiswa = new DataTable();
    da.Fill(dtMahasiswa);
    return dtMahasiswa;
}

1 reference
public DataTable getDataChartByTahun(DateTime thMasuk)
{
    if (conn.State == ConnectionState.Closed)
    {
        conn.Open();
    }

    SqlCommand cmd = new SqlCommand("sp_DashBoardByTahun", conn);
    cmd.CommandType = CommandType.StoredProcedure;
    cmd.Parameters.AddWithValue("@inTglMsuk", thMasuk.Year);
    da = new SqlDataAdapter(cmd);
    dtMahasiswa = new DataTable();
    da.Fill(dtMahasiswa);
    return dtMahasiswa;
}

```

7. Buat Form Baru untuk Dashboard



8. Tambahkan Code berikut pada class Form Dashboard

```
DAL dbLogic = new DAL();
bool isInitializing = true;
DataTable dt;
int button = 0;
```

9. Tambahkan code pada constuktor dashboard.

```
dtpTanggalMasuk.MinDate = new DateTime(2000, 1, 1);
dtpTanggalMasuk.Format = DateTimePickerFormat.Custom;
dtpTanggalMasuk.CustomFormat = "yyyy";
dtpTanggalMasuk.ShowUpDown = true;
dtpTanggalMasuk.MaxDate = DateTime.Now;

cmbTipe.DropDownStyle = ComboBoxStyle.DropDownList;
var items = new List<KeyValuePair<string, SeriesChartType>>
{
    new KeyValuePair<string, SeriesChartType>("Kolom", SeriesChartType.Column),
    new KeyValuePair<string, SeriesChartType>("Pie", SeriesChartType.Pie)
};

isInitializing = true;

cmbTipe.DataSource = items;
cmbTipe.DisplayMember = "Key";
cmbTipe.ValueMember = "Value";
cmbTipe.SelectedIndex = 0;

isInitializing = false;
loadDataChart();
```

10. Tambahkan method loadChartData()

```
public void loadChartData()
{
    chartProdi.Series.Clear();
    chartProdi.Titles.Clear();
    chartProdi.Legends.Clear();
    chartProdi.ChartAreas.Clear();

    ChartArea ca = new ChartArea("MainArea");
    ca.AxisX.Title = "Program Studi";
    ca.AxisY.Title = "Jumlah Mahasiswa";
    ca.AxisX.LabelStyle.Angle = -45;
    ca.BackColor = Color.Transparent;
    chartProdi.ChartAreas.Add(ca);
    try
    {
        if(button == 1)
        {
            dt = dbLogic.getDataChartByTahun(dtpTanggalMasuk.Value);
        }
        else
        {
            dt = dbLogic.getAllDataChart();
        }

        SeriesChartType tipe = (SeriesChartType)cmbTipe.SelectedValue;
        if (tipe == SeriesChartType.Column)
        {
            Series s = new Series("Mahasiswa");
            s.ChartType = SeriesChartType.Column;
            foreach (DataRow row in dt.Rows)
            {
                string prodi = row["NamaProdi"].ToString();
                int jumlah = Convert.ToInt32((long)row["JmlhMhs"]);
                s.Points.AddXY(prodi, jumlah);
            }
            chartProdi.Series.Add(s);
        }
        else
        {
            Series s = new Series("Jumlah Mahasiswa");
            s.ChartType = tipe;

            s.IsValueShownAsLabel = true;
            s.Label = "#VAL";
            s.LegendText = "#VALX";

            foreach (DataRow row in dt.Rows)
            {
                string prodi = row["NamaProdi"].ToString();
                int jumlah = Convert.ToInt32((long)row["JmlhMhs"]);

                s.Points.AddXY(prodi, jumlah);
            }
            chartProdi.Series.Add(s);
        }
    }
}
```

```

    catch (Exception ex)
    {
        MessageBox.Show("Gagal load data: " + ex.Message);
    }

    Title title = new Title("Jumlah Mahasiswa per Program Studi", Docking.Top, new Font("Arial", 14, FontStyle.Bold), Color.DarkBlue);
    chartProdi.Titles.Add(title);
    Legend legend = new Legend("MainLegend");
    legend.Docking = Docking.Right;
    chartProdi.Legends.Add(legend);
}

```

11. Tambahkan Event SelectedValueChanged pada cmbTipe, dan isi event dengan code berikut.

```

if (isInitializing)
    return;

if (button == 1) {
}
else {
    loadDataChart();
}

```

12. Tambahkan Event Klik pada button, dan isi event dengan code berikut.

- a. Button Load

```

button = 1;
loadDataChart();

```

- b. Button Reset

```

button = 0;
loadDataChart();

```

- c. Button Data Mahasiswa

```

DataMahasiswa frm1 = new DataMahasiswa();
frm1.Show();
this.Hide();

```

13. Ubah Design Halaman Data Mahasiswa

The screenshot shows a Windows application window titled "Data Mahasiswa". The window has a standard Windows title bar with minimize, maximize, and close buttons. Below the title bar is a toolbar with navigation icons (back, forward, search, etc.) and a status bar at the bottom showing "0". The main area of the window contains a form with the following fields and controls:

- NIM:** A text input field with a "Cari" button next to it.
- Nama:** A text input field.
- Jenis Kelamin:** A dropdown menu.
- Tanggal Lahir:** A date picker showing "16/ 6/2026".
- Alamat:** A text input field.
- Kode Prodi:** A text input field.
- Buttons:** A collection of buttons on the right side: "Open", "Reset Data" (green), "Insert", "Test" (red), "Update", "Rekap Data", "Delete", "Import form Excel", "Load", "Import to Database", "Refresh", and "Upload Gambar" (located below a dashed rectangular box).
- Data List:** A large, empty rectangular area at the bottom, likely intended for displaying a list of student data.

NB :

Atas button upload gambar adalah PictureBox.

14. Ubah Code pada beberapa Method dan Event dengan memanfaatkan Class DAL.

- Tambahkan object untuk mendeklasasikan class DAL.

```
DAL dbLogic = new DAL();
```


- b. Ubah code pada Method HitungTotal, LoadData, ClearForm, dan simpanLog

```
private void HitungTotal()
{
    try
    {
        int total = (dbLogic.CountMhs().Equals(DBNull.Value)) ? 0 : dbLogic.CountMhs();

        lblCountMhs.Text = "Total Mahasiswa : " + total;
    }
    catch (Exception ex)
    {
        simpanLog(ex.Message);
        MessageBox.Show("Gagal load data: " + ex.Message);
    }
}
```

```
private void LoadData()
{
    try
    {
        bindingSource1.DataSource = dbLogic.GetMhs();
        dataGridView1.DataSource = bindingSource1;
        DataGridViewImageColumn fotoColumn = (DataGridViewImageColumn)dataGridView1.Columns["Foto"];
        fotoColumn.ImageLayout = DataGridViewImageCellLayout.Stretch;

        HitungTotal();
        foreach (DataGridViewColumn col in dataGridView1.Columns)
        {
            Console.WriteLine("Name: " + col.Name + " | DataPropertyName: " + col.DataPropertyName);
        }
        dataGridView1.Enabled = true;
        btnImpDb.Enabled = false;
        btnInsert.Enabled = true;
        btnUpdate.Enabled = true;
        btnDelete.Enabled = true;
        btnCari.Enabled = true;
        btnLoad.Enabled = true;
        btnReset.Enabled = true;
        btnTestInjection.Enabled = true;
    }
    catch (Exception ex)
    {
        simpanLog(ex.Message);
        MessageBox.Show("Gagal load data: " + ex.Message);
    }
}
```

```
private void ClearForm()
{
    txtNIM.Enabled = true;
    txtNIM.Clear();
    txtNama.Clear();
    cmbJK.SelectedIndex = -1;
    txtAlamat.Clear();
    txtKodeProdi.Clear();
    dtpTanggalLahir.Value = DateTime.Now;
    fotoMhs.Image = null;
    txtNIM.Focus();
}
```

```
public void simpanLog(string message)
{
    dbLogic.InsertLog(message);
}
```

c. Ubah code pada Event Click

```
private void btnConnect_Click(object sender, EventArgs e)
{
    try
    {
        using (MySqlConnection conn = new MySqlConnection(dbLogic.GetConnectionString()))
        {
            conn.Open();
            MessageBox.Show("Koneksi Berhasil");
        }
    }
    catch (SQLException ex)
    {
        simpanLog(ex.Message);
        MessageBox.Show("SQL Error :" + ex.Message);
    }
    catch (Exception ex)
    {
        simpanLog(ex.Message);
        MessageBox.Show("General Error :" + ex.Message);
    }
}
```

```

private void btnInsert_Click(object sender, EventArgs e)
{
    try
    {
        byte[] ConvertImageToBytes(PictureBox pb)
        {
            using (MemoryStream ms = new MemoryStream())
            {
                pb.Image.Save(ms, System.Drawing.Imaging.ImageFormat.Jpeg);
                return ms.ToArray();
            }
        }
        byte[] imgBytes = ConvertImageToBytes(fotoMhs);
        dbLogic.InsertMhs(txtNIM.Text, txtNama.Text, txtAlamat.Text, cmbJK.Text, dtpTanggallahir.Value.Date, txtKodeProdi.Text, imgBytes);
        MessageBox.Show("Data mahasiswa berhasil ditambahkan");
        ClearForm();
        LoadData();
    }
    catch (SqlException ex)
    {
        simpanLog("Rollback Insert : " + ex.Message);
        MessageBox.Show("SQL Error : " + ex.Message);
    }
    catch (Exception ex)
    {
        simpanLog("General Error : " + ex.Message);
        MessageBox.Show("General Error : " + ex.Message);
    }
}

```

```

private void btnUpdate_Click(object sender, EventArgs e)
{
    try
    {
        byte[] ConvertImageToBytes(PictureBox pb)
        {
            using (MemoryStream ms = new MemoryStream())
            {
                pb.Image.Save(ms, System.Drawing.Imaging.ImageFormat.Jpeg);
                return ms.ToArray();
            }
        }
        byte[] imgBytes = ConvertImageToBytes(fotoMhs);
        dbLogic.UpdateMhs(txtNIM.Text, txtNama.Text, txtAlamat.Text, cmbJK.Text, dtpTanggallahir.Value.Date, txtKodeProdi.Text, imgBytes);
        MessageBox.Show("Data mahasiswa berhasil diubah");
        ClearForm();
        btnLoad.PerformClick();
    }
    catch (SqlException ex)
    {
        simpanLog(ex.Message);
        MessageBox.Show("SQL Error : " + ex.Message);
    }
    catch (Exception ex)
    {
        simpanLog(ex.Message);
        MessageBox.Show("General Error : " + ex.Message);
    }
}

```

```

private void btnDelete_Click(object sender, EventArgs e)
{
    try
    {
        DialogResult dg = MessageBox.Show(
            "Yakin ingin menghapus data?",
            "Konfirmasi",
            MessageBoxButtons.YesNo,
            MessageBoxIcon.Question);
        if (dg == DialogResult.Yes)
        {
            dbLogic.DeleteMhs(txtNIM.Text);
            MessageBox.Show("Data mahasiswa berhasil dihapus");
            ClearForm();
            btnLoad.PerformClick();
        }
    }
    catch (SqlException ex)
    {
        simpanLog(ex.Message);
        MessageBox.Show("SQL Error : " + ex.Message);
    }
    catch (Exception ex)
    {
        simpanLog(ex.Message);
        MessageBox.Show("General Error : " + ex.Message);
    }
}

```

```

private void btnReset_Click(object sender, EventArgs e)
{
    try
    {
        dbLogic.resetData();
        MessageBox.Show("Data berhasil direset");
        LoadData();
    }
    catch (SqlException ex)
    {
        simpanLog(ex.Message);
        MessageBox.Show("SQL Error : " + ex.Message);
    }
    catch (Exception ex)
    {
        simpanLog(ex.Message);
        MessageBox.Show("General Error : " + ex.Message);
    }
}

```

```

private void btnTestInjection_Click(object sender, EventArgs e)
{
    try
    {
        dbLogic.testInject(txtNIM.Text);

        LoadData();
    }
    catch (SQLException ex)
    {
        if(ex.Message.Contains("safe"))
        {
            simpanLog(ex.Message);
            MessageBox.Show("SQL Error : Unsafe UPDATE operation not allowed");
        }
        else
        {
            simpanLog(ex.Message);
            MessageBox.Show("SQL Error :" + ex.Message);
        }
    }
    catch (Exception ex)
    {
        simpanLog(ex.Message);
        MessageBox.Show("General Error :" + ex.Message);
    }
}

```

d. Ubah code pada Event DataGridView

```

private void dataGridView1_CellClick(object sender, DataGridViewCellEventArgs e)
{
    if (e.RowIndex >= 0)
    {
        DataRow row = ((DataRowView)bindingSource1[e.RowIndex]).Row;
        txtNIM.Text = row[0].ToString();
        txtNama.Text = row[1].ToString();
        cmbJK.Text = row[2].ToString();
        dtpTanggallahir.Value = Convert.ToDateTime(row[3]);
        txtAlamat.Text = row[4].ToString();
        txtKodeProdi.Text = row[6].ToString();

        if (row[5] != DBNull.Value)
        {
            byte[] imgBytes = (byte[])row[5];
            using (MemoryStream ms = new MemoryStream(imgBytes))
            {
                fotoMhs.Image = Image.FromStream(ms);
                fotoMhs.SizeMode = PictureBoxSizeMode.StretchImage;
            }
        }
        else
        {
            fotoMhs.Image = null;
        }

        txtNIM.Enabled = false;
    }
}

```

- e. Tambahkan event Klik, pada Button yang baru ditambahkan ke dalam Form. Kemudian tambahkan code berikut untuk masing-masing event.

```
private void btnUpload_Click(object sender, EventArgs e)
{
    OpenFileDialog ofd = new OpenFileDialog();
    ofd.Filter = "Image Files|*.jpg;*.jpeg;*.png;*.bmp";

    if (ofd.ShowDialog() == DialogResult.OK)
    {
        fotoMhs.Image = Image.FromFile(ofd.FileName);
        fotoMhs.SizeMode = PictureBoxSizeMode.StretchImage;
    }
}
```

```
private void btnImpExcel_Click(object sender, EventArgs e)
{
    using (OpenFileDialog openFileDialog = new OpenFileDialog { Filter = "Excel Workbook| *.xlsx" })
    {
        if (openFileDialog.ShowDialog() == DialogResult.OK)
        {
            string filePath = openFileDialog.FileName;
            using (var stream = File.Open(filePath, FileMode.Open, FileAccess.Read))
            {
                using (var reader = ExcelReaderFactory.CreateReader(stream))
                {
                    var result = reader.AsDataSet(new ExcelDataSetConfiguration()
                    {
                        ConfigureDataTable = (_) => new ExcelDataTableConfiguration()
                        {
                            UseHeaderRow = true
                        }
                    });
                    DataTable dt = result.Tables[0];
                    dataGridView1.DataSource = dt;
                    dataGridView1.Enabled = false;

                    btnImpDb.Enabled = true;
                    btnInsert.Enabled = false;
                    btnUpdate.Enabled = false;
                    btnDelete.Enabled = false;
                    btnCari.Enabled = false;
                    btnLoad.Enabled = false;
                    btnReset.Enabled = false;
                    btnTestInjection.Enabled = false;
                }
            }
        }
    }
}
```



```

private void btnImpDb_Click(object sender, EventArgs e)
{
    try
    {
        DataTable dt = (DataTable)dataGridView1.DataSource;

        if (dt == null || dt.Rows.Count == 0)
        {
            MessageBox.Show("Tidak ada data untuk diimport.");
            return;
        }

        int sukses = 0;

        foreach (DataRow row in dt.Rows)
        {
            string nim = row["NIM"].ToString().Trim();
            string nama = row["Nama"].ToString().Trim();
            string jk = row["JenisKelamin"].ToString().Trim();
            string alamat = row["Alamat"].ToString().Trim();
            string kodeProdi = row["NamaProdi"].ToString().Trim();
            string fotoPath = row.Table.Columns.Contains("FotoPath")
                ? row["FotoPath"].ToString().Trim()
                : string.Empty;

            if (string.IsNullOrEmpty(nim) || string.IsNullOrEmpty(nama))
                continue;

            DateTime tglLahir;
            if (!DateTime.TryParse(row["TanggalLahir"].ToString(), out tglLahir))
                continue;
        }
    }
}

```

```

byte[] ConvertImageFromPath(string path)
{
    if (string.IsNullOrEmpty(path))
        return null;

    if (!File.Exists(path))
        return null;

    return File.ReadAllBytes(path);
}
byte[] fotoBytes = ConvertImageFromPath(fotoPath);

dbLogic.InsertMhs(nim, nama, alamat, jk, tglLahir, kodeProdi, fotoBytes);

sukses++;
}

MessageBox.Show("Data mahasiswa berhasil ditambahkan");
ClearForm();
LoadData();
}
catch (SQLException ex)
{
    simpanLog("Rollback Insert :" + ex.Message);
    MessageBox.Show("SQL Error :" + ex.Message);
}
catch (Exception ex)
{
    simpanLog("General Error :" + ex.Message);
    MessageBox.Show("General Error :" + ex.Message);
}
}
}

```

15. Tambahkan object untuk mendeklasasikan class DAL pada Form RekapMahasiswa

```
DAL dbLogic = new DAL();
```

16. Ubah Konstruktor Report

```
public Report(string Prodi, DateTime TglMasuk)
{
    InitializeComponent();

    prodi = Prodi;
    tglmasuk = TglMasuk;

    try
    {
        DataTable dtMahasiswa = dbLogic.getDataRekap(prodi, tglmasuk);

        listMahasiswa.SetDataSource(dtMahasiswa);
        crystalReportViewer1.ReportSource = listMahasiswa;
        crystalReportViewer1.Refresh();
    }
    catch (Exception ex)
    {
        //simpanLog(ex.Message);
        MessageBox.Show("Gagal load data: " + ex.Message);
    }
}
```

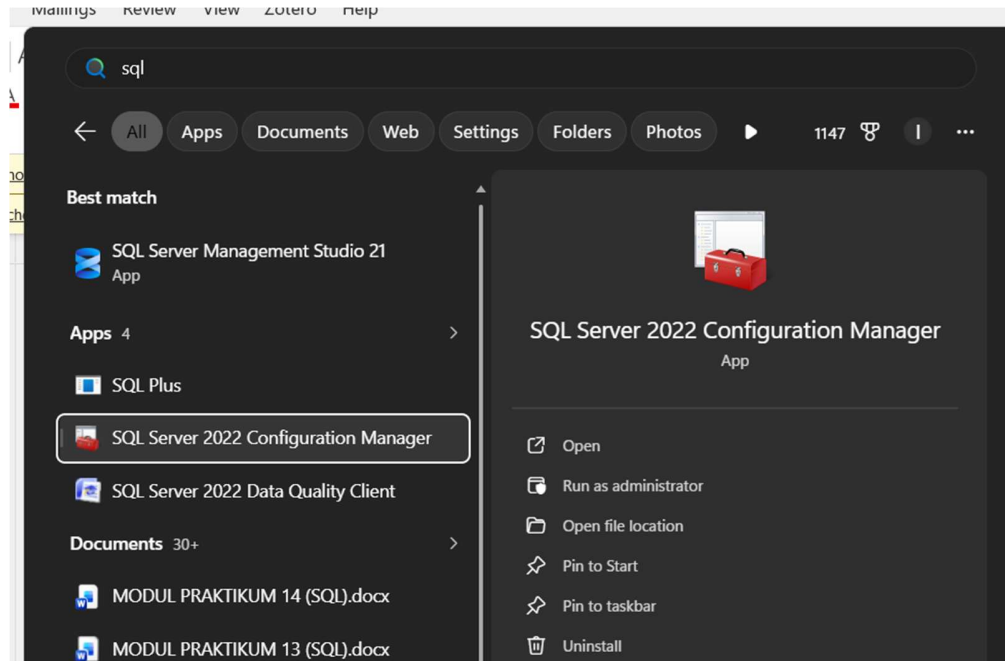
17. Ubah Class Program, agar ketika diJalankan Form yang pertama kali muncul adalah Form Dashboard

```
internal static class Program
{
    /// <summary>
    /// The main entry point for the application.
    /// </summary>
    [STAThread]
    0 references
    static void Main()
    {
        Application.EnableVisualStyles();
        Application.SetCompatibleTextRenderingDefault(false);
        Application.Run(new Dashboard());
    }
}
```

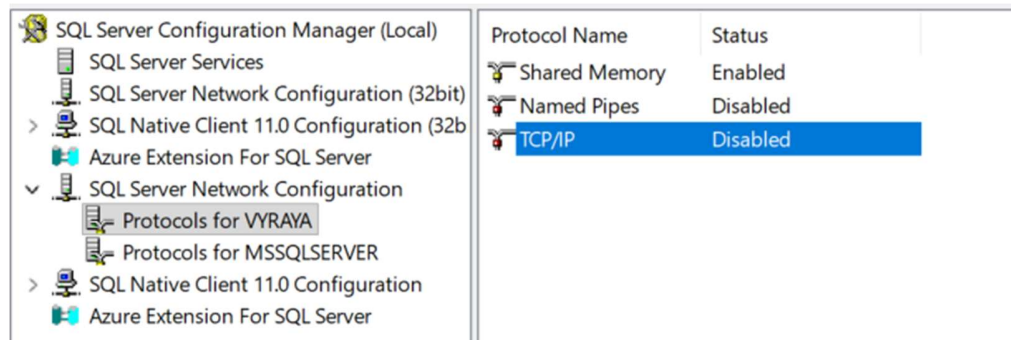
Deploy Aplikasi

18. Cofigurasi Server SQL agar bisa diakses secara public dalam satu jaringan yang sama.

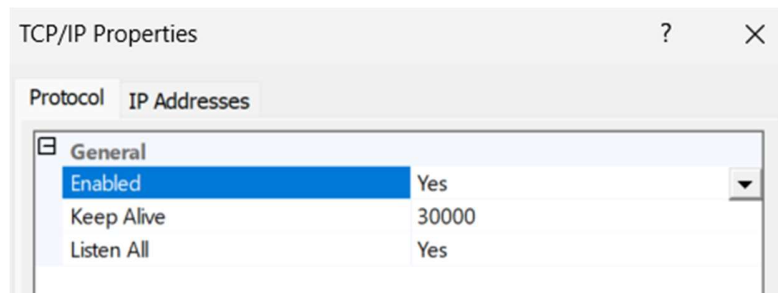
- a. Buka SQL Sever Configuration Manager



- b. Pada bagian SQL Server Configuration, double klik TCP/IP Server.



- c. Ubah protocol, pada bagian enable dari No ke Yes, kemudian Apply dan Ok



19. Menggubah konfigurasi string koneksi pada SqlConnection :

- a. Tambahkan Function untuk mengambil IPAddress Server pada class DAL

```
public static string GetLocalIPAddress()
{
    string localIP = string.Empty;
    try
    {
        var host = System.Net.Dns.GetHostEntry(System.Net.Dns.GetHostName());
        foreach (var ip in host.AddressList)
        {
            if (ip.AddressFamily == System.Net.Sockets.AddressFamily.InterNetwork)
            {
                localIP = ip.ToString();
                break;
            }
        }
    }
    catch (Exception ex)
    {
        MessageBox.Show("Error getting local IP address: " + ex.Message);
    }
    return localIP;
}
```

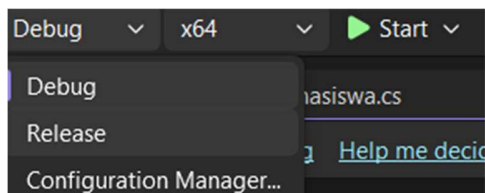
- b. Ubah string connection pada class DAL

```
public static string GetConnectionString()
{
    string connectionString = $"Data Source={GetLocalIPAddress()};Initial Catalog=DBAkademikADO;User ID=sa;Password=PasswordSA;";
    return connectionString;
}

SqlConnection conn = new SqlConnection(GetConnectionString());
```

20. Membuat release aplikasi

- a. Ganti mode debug menjadi release



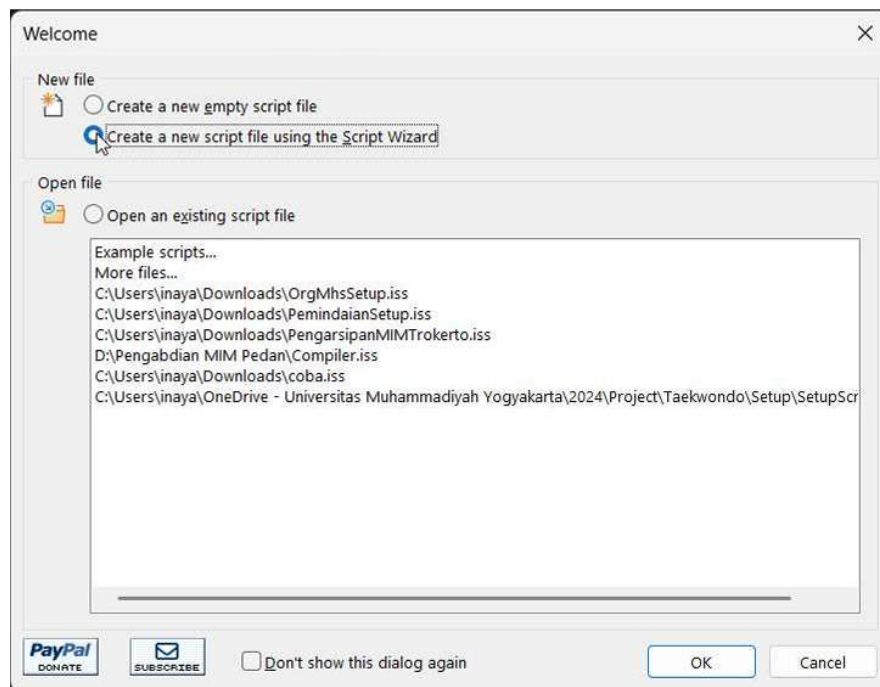
- b. Build solution
c. Cek pada folder project hasil release aplikasi

- Lokasi path : [localdisk]\[namaproject]\[namaproject]\bin\release
- Contoh :

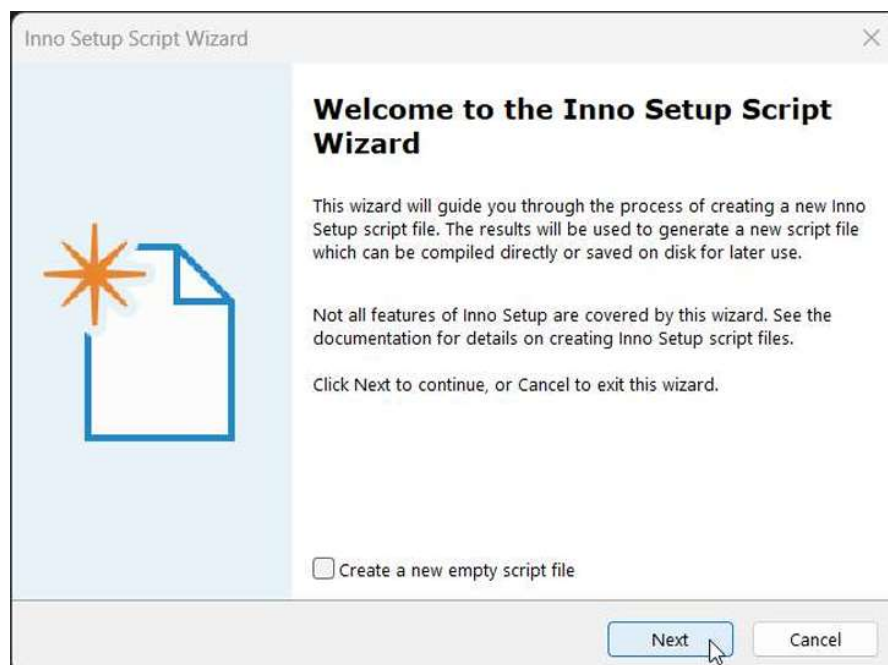
OneDrive > ... TI UMY > PABD 2025 > Code > DataSet_DataAdapter > DataSet_DataAdapter > bin > Release

21. Membuat setup aplikasi menggunakan Inno Setup Compiler

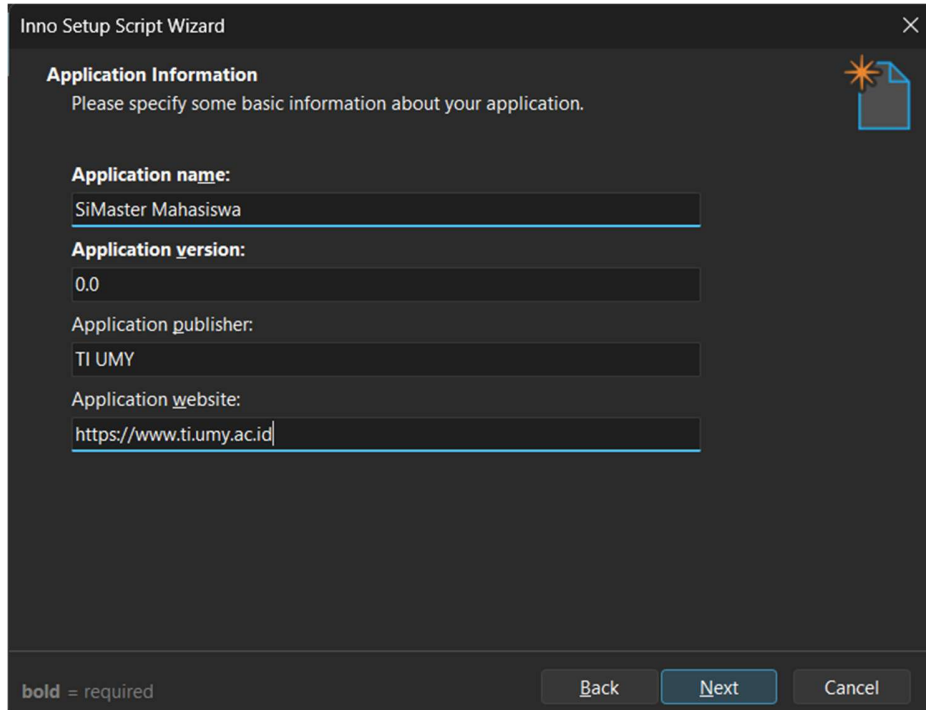
- a. Buka Inno Setup Compiler
- b. Pilih “create a new script file using the Script Wizard”, kemudian klik ok



- c. Klik next

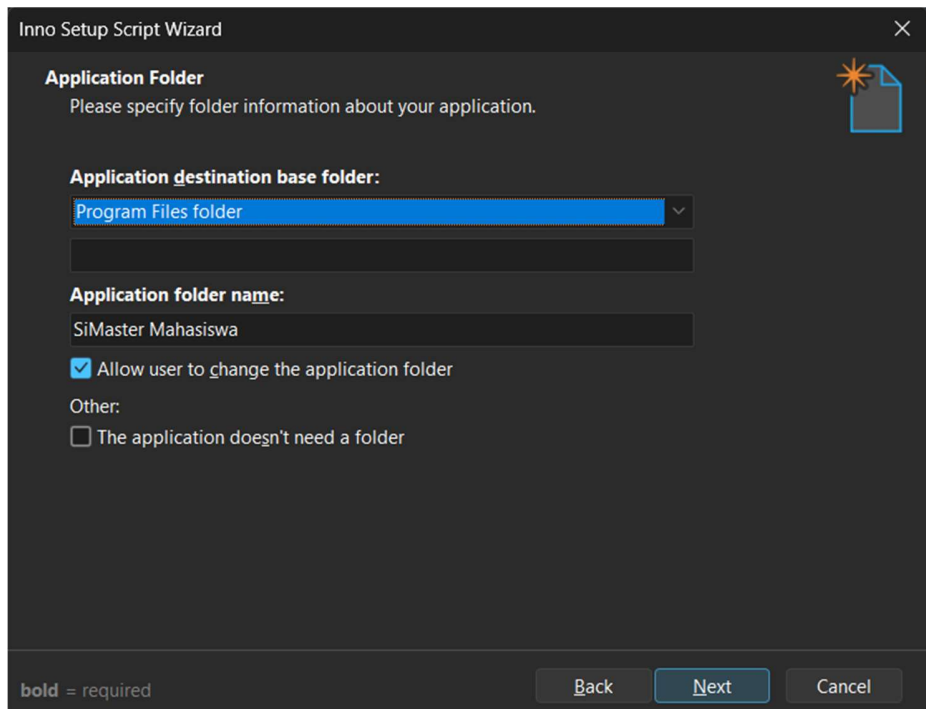


- d. Lengkapi informasi terkait aplikasi yang akan diinstal, untuk pengisian bebas, kemudian klik next



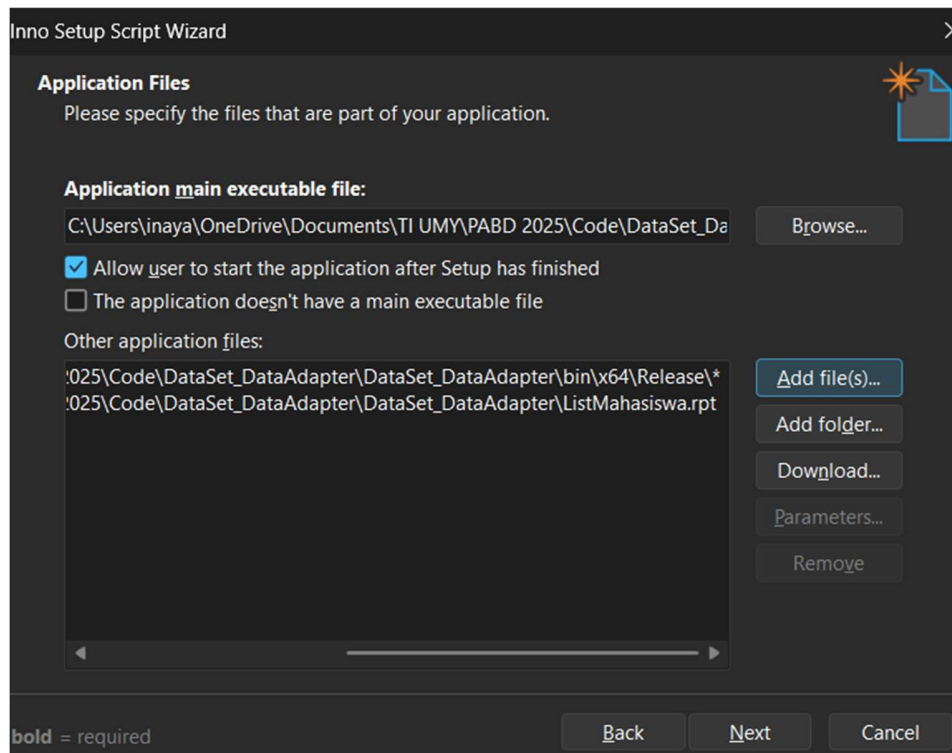
The screenshot shows the 'Inno Setup Script Wizard' window, specifically the 'Application Information' step. The window has a dark theme and a close button (X) in the top right corner. Below the title bar, there is a sub-header 'Application Information' and a prompt 'Please specify some basic information about your application.' To the right of the prompt is a small icon of a document with a star. The form contains five text input fields: 'Application name:' with the value 'SiMaster Mahasiswa', 'Application version:' with the value '0.0', 'Application publisher:' with the value 'TI UMY', 'Application website:' with the value 'https://www.ti.umy.ac.id', and a 'bold = required' label at the bottom left. At the bottom right, there are three buttons: 'Back', 'Next' (highlighted in blue), and 'Cancel'.

- e. Tentukan lokasi penginstalan aplikasi dan apakah pengguna boleh merubah lokasi penginstalan. Jika ingin menginstal pada folder program file dan pengguna diizinkan melakukan perubahan lokasi penginsatalan, klik next

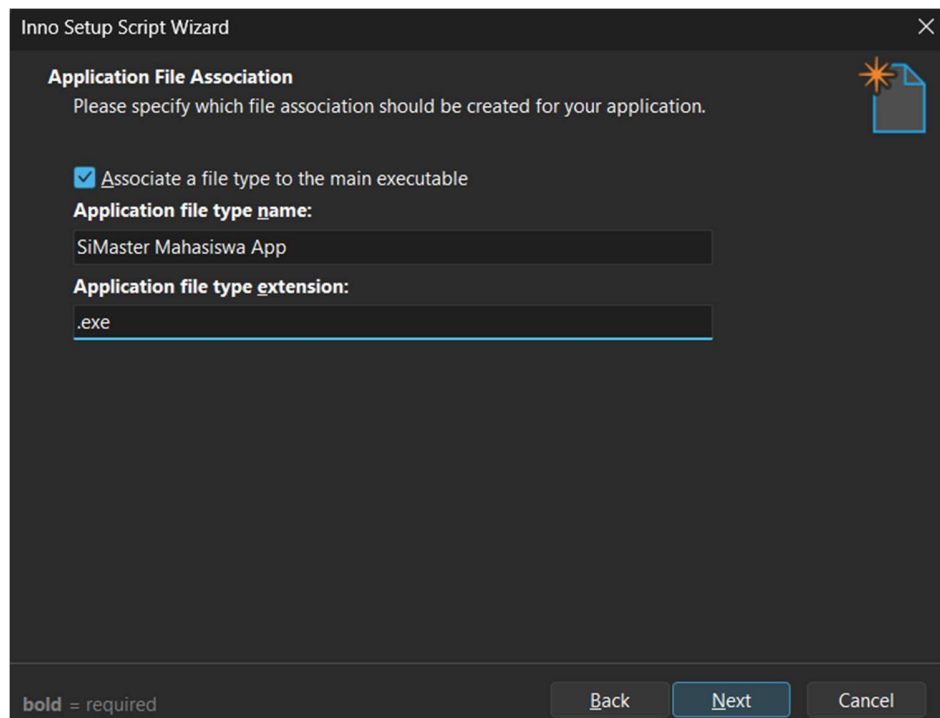


The screenshot shows the 'Inno Setup Script Wizard' window, specifically the 'Application Folder' step. The window has a dark theme and a close button (X) in the top right corner. Below the title bar, there is a sub-header 'Application Folder' and a prompt 'Please specify folder information about your application.' To the right of the prompt is a small icon of a document with a star. The form contains a dropdown menu for 'Application destination base folder:' with 'Program Files folder' selected, a text input field for 'Application folder name:' with the value 'SiMaster Mahasiswa', a checked checkbox for 'Allow user to change the application folder', and an 'Other:' section with an unchecked checkbox for 'The application doesn't need a folder'. At the bottom left, there is a 'bold = required' label. At the bottom right, there are three buttons: 'Back', 'Next' (highlighted in blue), and 'Cancel'.

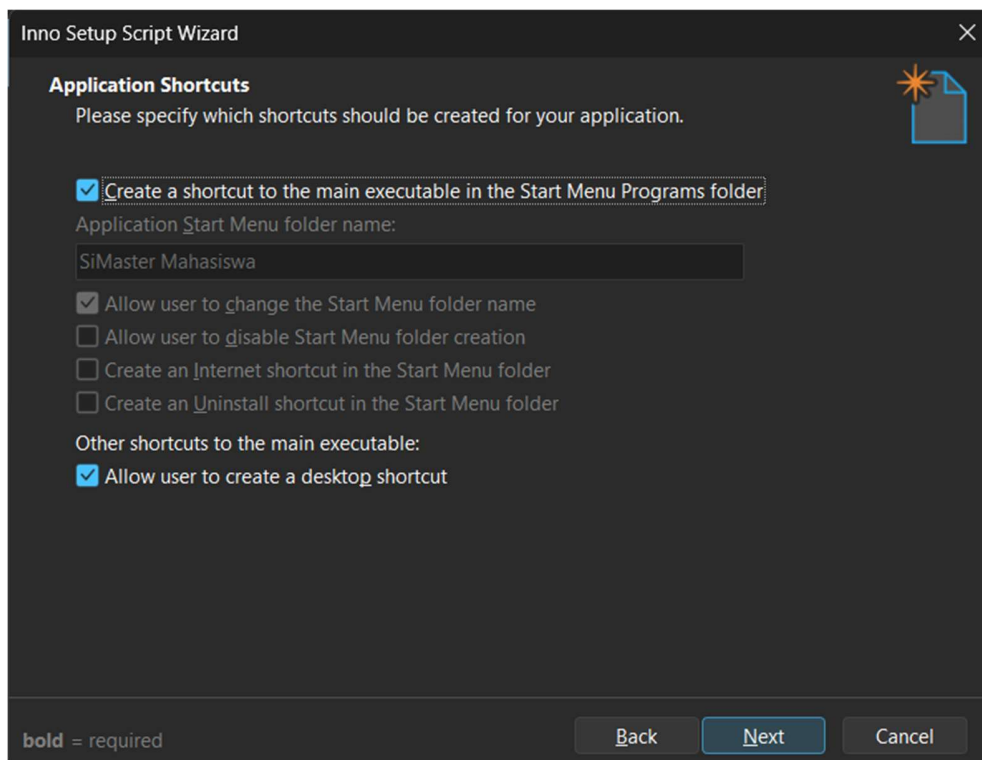
- f. Pilih file hasil release aplikasi, untuk menjadi main aplikasi, dan tambahkan folder release serta file design reportviewer, agar seluruh package yang dibutuhkan aplikasi diinstal ke dalam folder instalasi aplikasi.



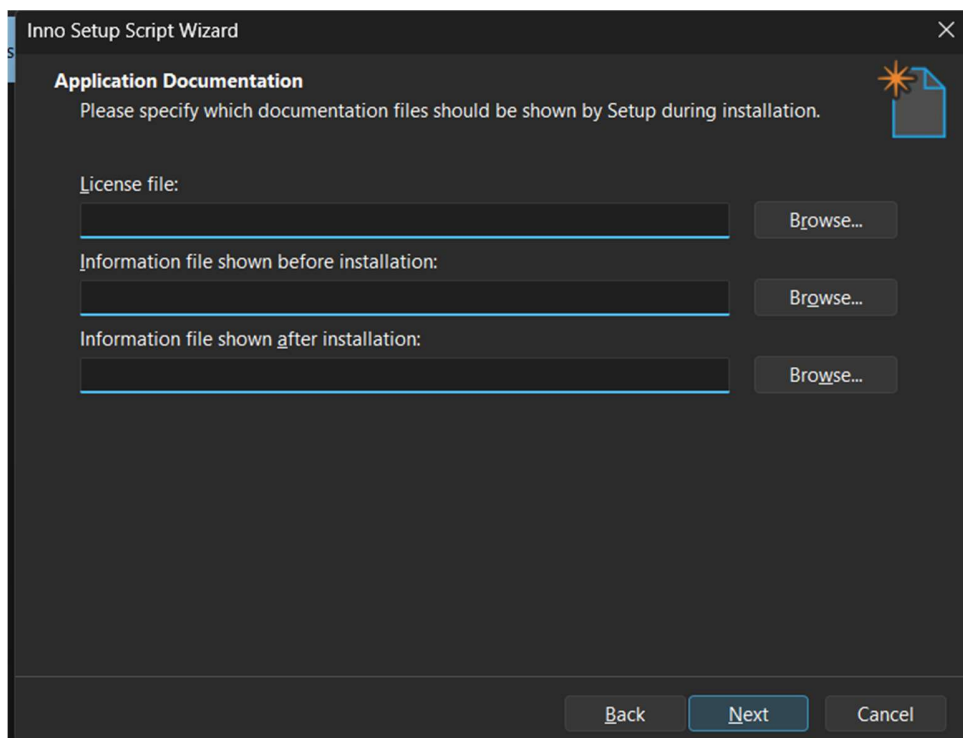
- g. Tentukan nama aplikasi yang akan dibuat



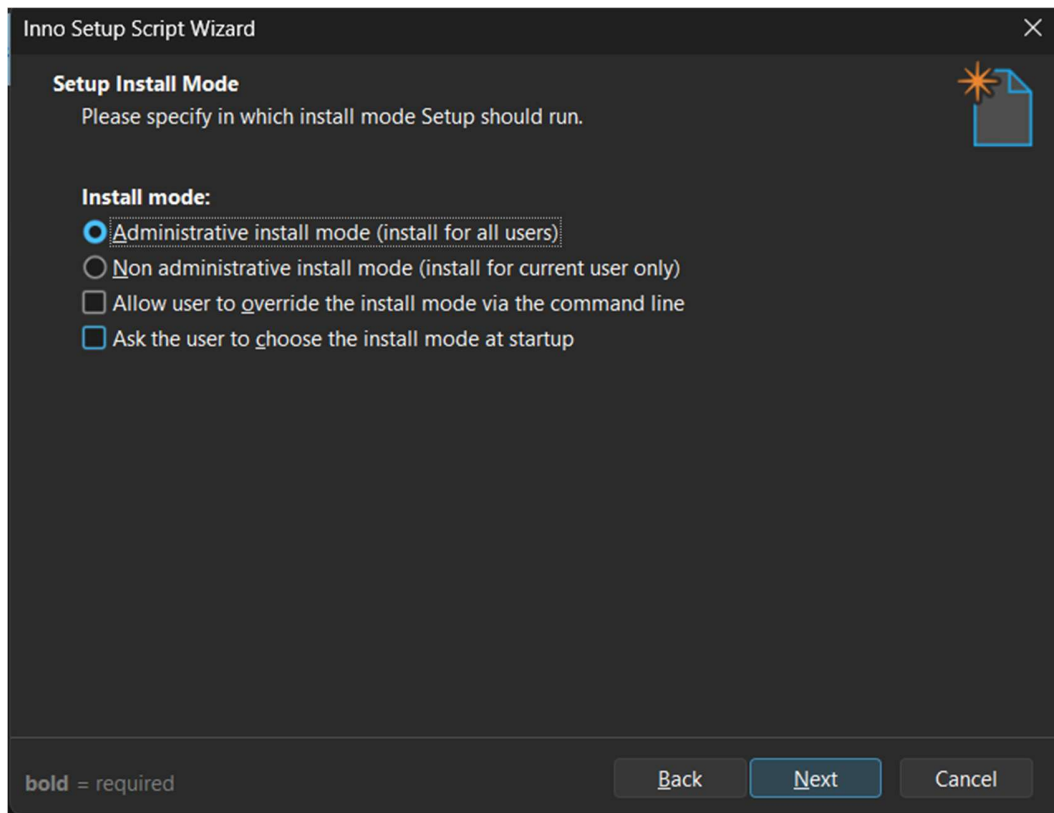
- h. Konfigurasi apakah user nanti dapat membuat file shortcuts aplikasi atau tidak



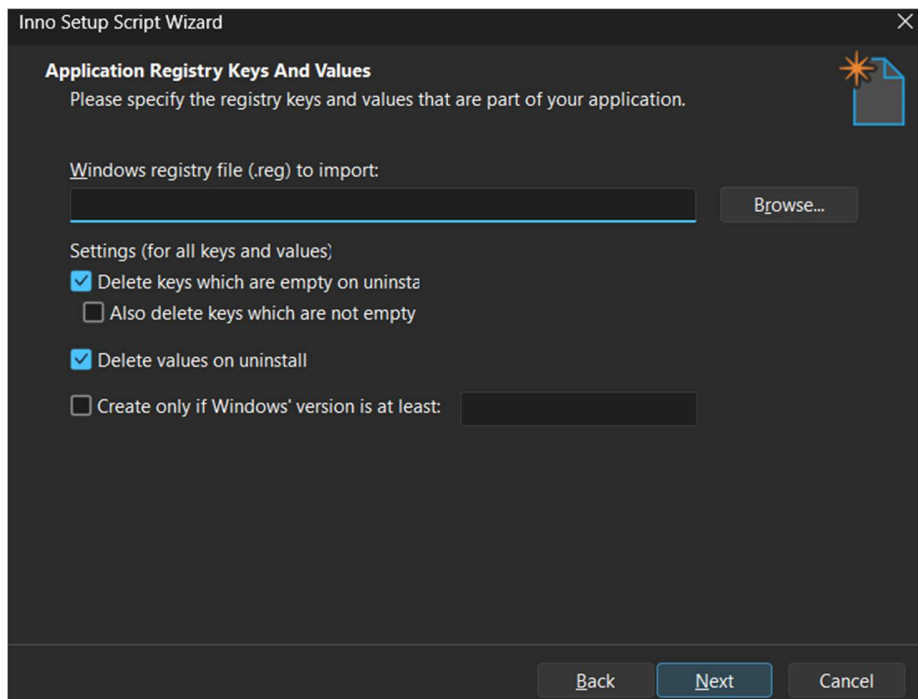
- i. Jika ada sertifikat yang harus diinstal pada aplikasi silahkan tambahkan pada bagian application documentation



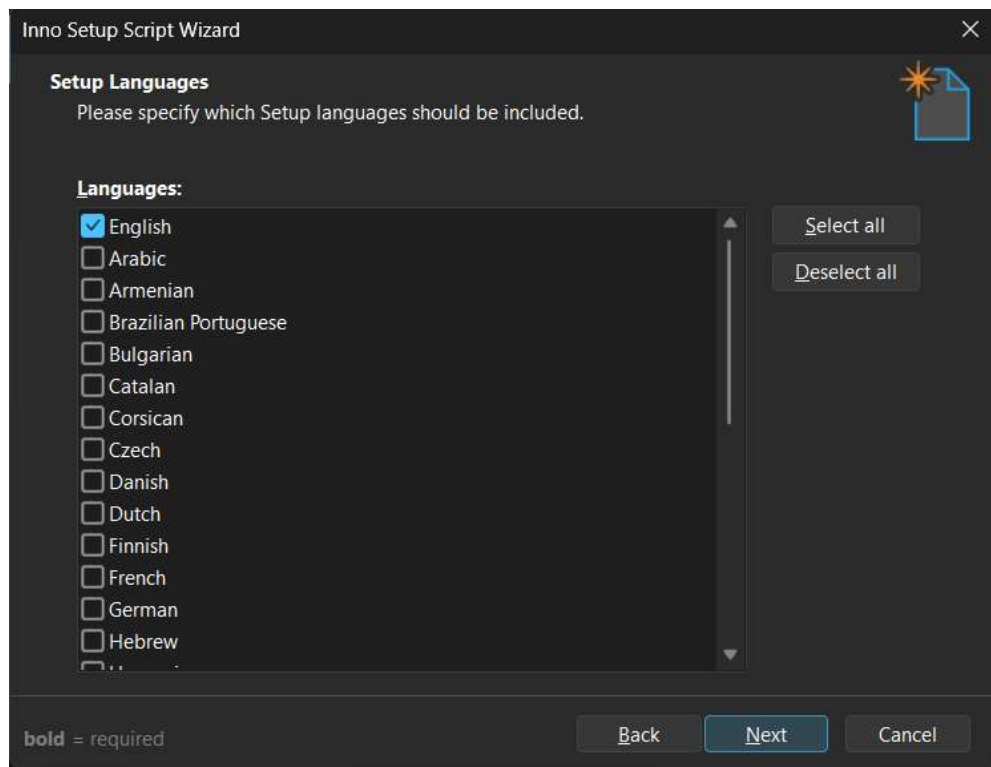
- j. Tentukan mode instalasi yang akan dilakukan pada setup aplikasi kalian



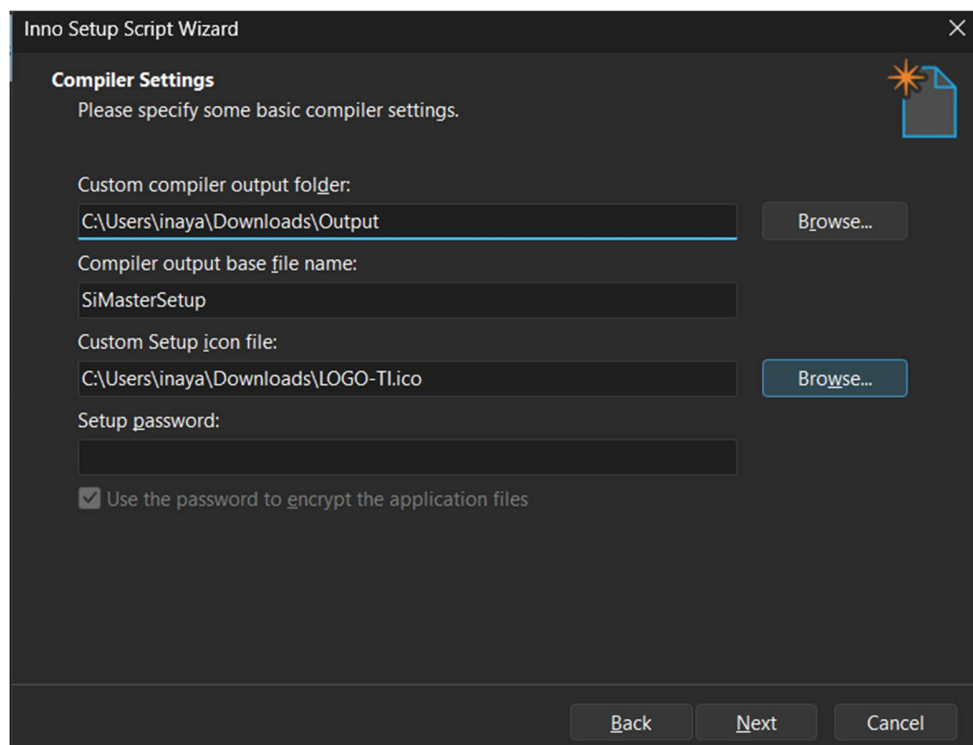
- k. Jika kalian memiliki registry keys pada aplikasi yang akan dideploy, silahkan menambahkan apa bagian berikut, dan atur proses uninstal aplikasi kalian pada bagian berikut



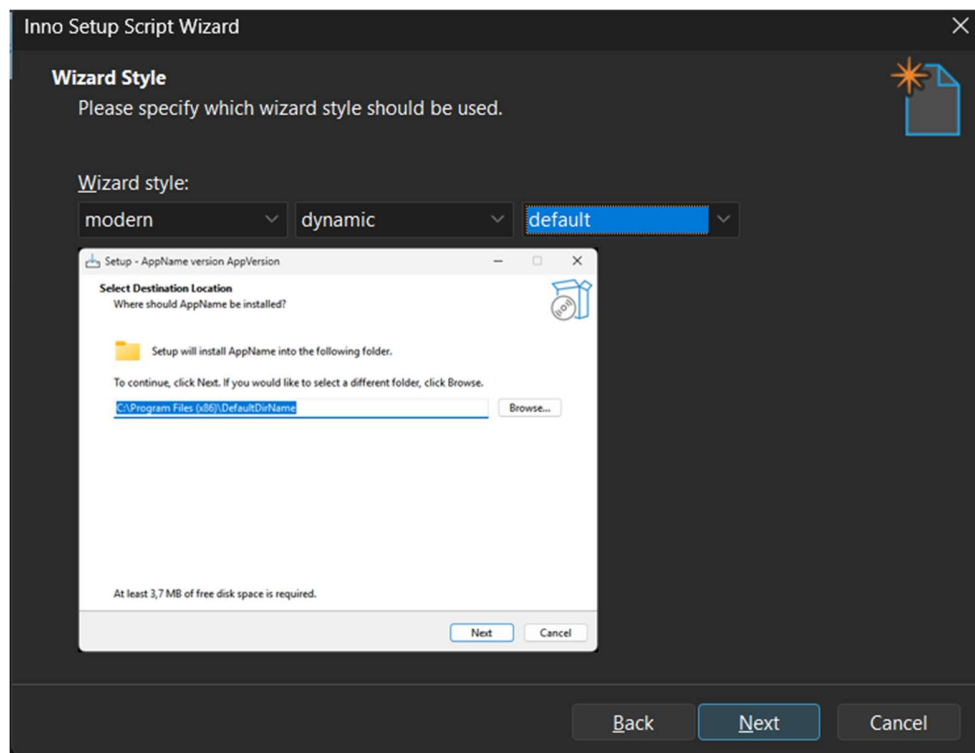
- l. Pilih bahasa apa saja yang dapat digunakan pada proses instalasi



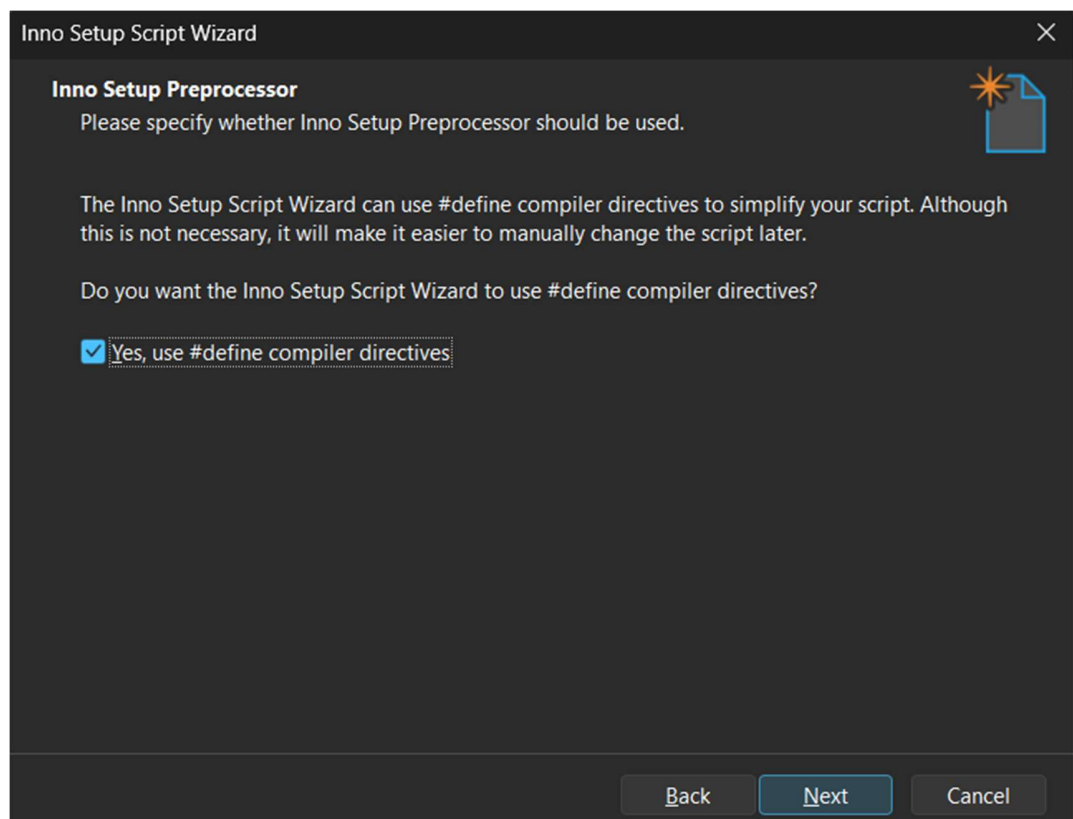
- m. Konfigurasi lokasi, nama, dan icon untuk membuat setup aplikasi, jika tidak ada yang ingin diubah klik next



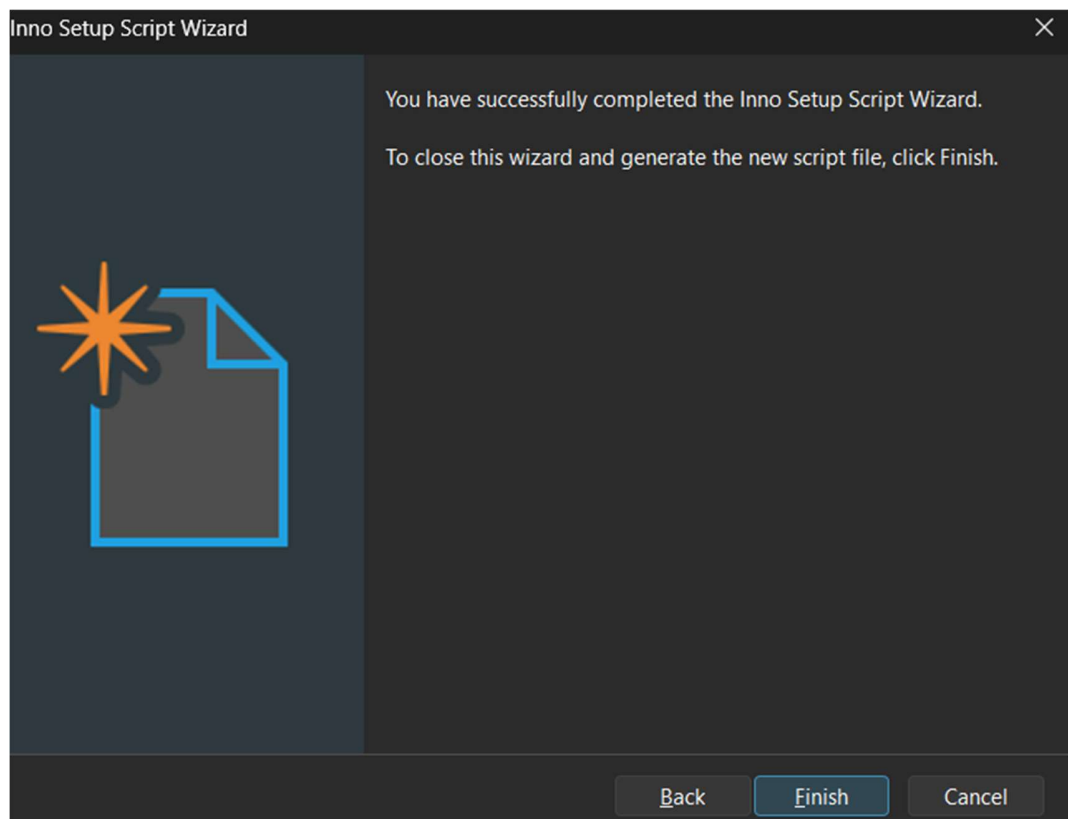
- n. Pilih mode tampilan setup



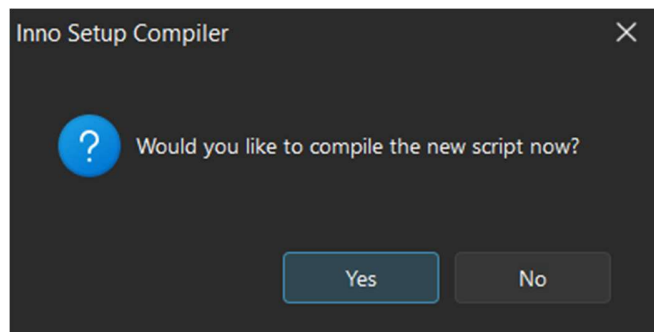
- o. Klik next pada bagian Inno Setup Preprocess



- p. Klik finish untuk melanjutkan proses pembuatan skrip setup



- q. Jika kalian ingin langsung membuat setup klik yes dan simpan file skrip yang telah dibuat. Jika kalian ingin mengubah skrip compiler, klik no

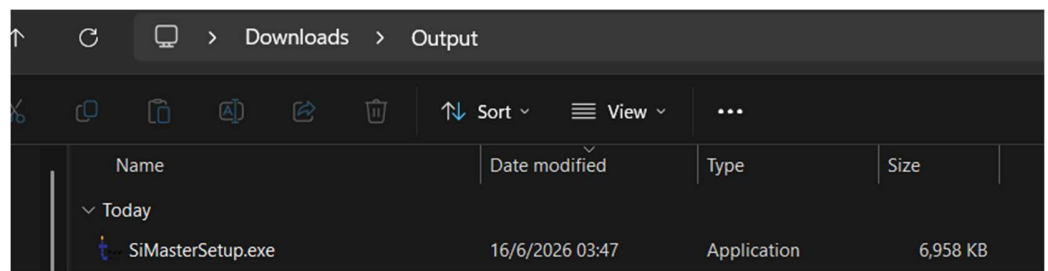


- r. Jika hasil compiler setup sudah selesai, silahkan buka folder lokasi setup dibuat
- Status compiler selesai

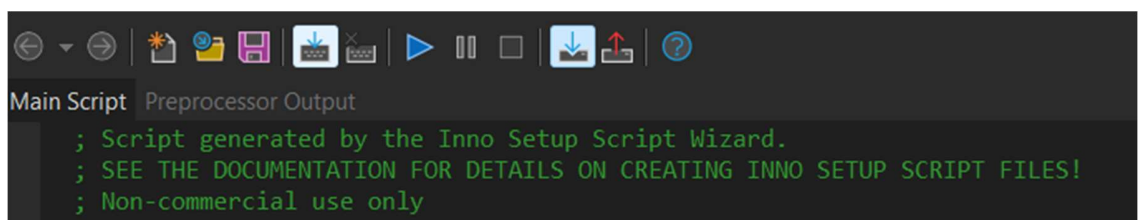
```
Compressing Setup program executable
Updating version info (Setup.exe)
Updating manifest (Setup.exe)

*** Finished. [3:47:57 am, 00:06.406 elapsed]
```

- Hasil file setup



- s. Jika kalian telah selesai mengubah skrip atau ada perubahan pada release aplikasi dan ingin menjalankan compiler setup, klik logo compiler



- t. Jalankan setup, jika sudah selesai buka folder tempat instalasi aplikasi, pastikan semua package telah terinstal dan file design reportviewer telah copy